

Fundamentals of Statistical Learning

Definitions cheatsheet

Definition 3.2: Random variable

A *random variable* is a measurable map

$$X : \Omega \longrightarrow \mathbb{R}.$$

It assigns one real number $X(\omega)$ to each outcome ω .

Definition 3.4: Preimage

For a random variable $X : \Omega \rightarrow \mathbb{R}$ and a set $E \subseteq \mathbb{R}$, the *preimage* (or *inverse image*) of E is

$$X^{-1}(E) := \{\omega \in \Omega : X(\omega) \in E\} \subseteq \Omega.$$

To find this event, we collect the outcomes whose outputs fall in E . Whenever $X^{-1}(E)$ is an event, we define

$$\begin{aligned} P(X \in E) &:= P(X^{-1}(E)) \\ &= P(\{\omega \in \Omega : X(\omega) \in E\}). \end{aligned}$$

In particular, for a single value x ,

$$P(X = x) := P(X^{-1}(\{x\})).$$

Here X is the random variable, while x is a particular real number.

Definition 3.5: Distribution of a random variable

The *distribution* of X is the rule that tells us with what probability X takes values in a given set E .

$$P_X(E) = P(X \in E).$$

Definition 3.6: Discrete random variable and PMF

A random variable X is *discrete* if there is a finite or countably infinite set^{*} $\mathcal{X} = \{x_1, x_2, \dots\}$ such that $P(X \in \mathcal{X}) = 1$.

Its *probability mass function* (PMF) is

$$p_X(x) := P(X = x), \quad x \in \mathbb{R}.$$

1. $p_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. The *support*, in the discrete sense used here, is $\text{supp}(X) := \{x \in \mathbb{R} : p_X(x) > 0\}$. Thus the support lists the values that have positive probability; values outside it have probability zero.
3. $\sum_{x \in \mathcal{X}} p_X(x) = 1$.

Definition 3.7: Continuous random variable and PDF

A random variable X is *continuous* (in the sense used here) if there exists a function f_X , called its *probability density function* (PDF), such that, for every $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

1. $f_X(x) \geq 0$ for every $x \in \mathbb{R}$.
2. Following the convention in the slides, the *support* is $\text{supp}(X) := \{x \in \mathbb{R} : f_X(x) > 0\}$.
3. $\int_{\mathbb{R}} f_X(x) dx = 1$.

Definition 3.8: Cumulative distribution function

Given a random variable X , its *cumulative distribution function* (CDF) is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(x) := P(X \leq x) = P(X \in (-\infty, x]), \quad x \in \mathbb{R}.$$

It gives the probability accumulated up to and including the threshold x . It is defined for both [discrete](#) and [continuous](#) random variables.

Definition 3.10: Quantile function: general case

Let X be any random variable, discrete or continuous, with CDF F_X . For a probability level $0 < p < 1$, its *quantile function* is

$$Q(p) := \inf\{x \in \mathbb{R} : F_X(x) \geq p\}.$$

It gives the smallest threshold x such that $P(X \leq x)$ is at least p . We also write q_p instead of $Q(p)$.

^{*}A finite set has finitely many elements. A set A is countably infinite if there exists a bijection $f : \mathbb{N} \rightarrow A$.

Relations

From the PMF to the CDF

What we are trying to answer

If we know the probability of each individual value of X , how do we calculate the probability that X is at most x ?

For a discrete random variable, the event $X \leq x$ includes every possible value k that is at most x . These events $\{X = k\}$ are pairwise disjoint, so we add their probabilities:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \sum_{\substack{k \in \text{supp}(X) \\ k \leq x}} \underbrace{p_X(k)}_{\text{PMF: known}}, \quad x \in \mathbb{R}.$$

From the PDF to the CDF

What we are trying to answer

We know how probability is spread over the possible values of X (its PDF). How do we calculate the probability of getting a value less than or equal to x ?

For a continuous random variable with PDF f_X , we calculate $P(X \leq x)$ by integrating the PDF over $(-\infty, x]$:

$$\underbrace{F_X(x)}_{\text{CDF: to find}} = P(X \leq x) = \int_{-\infty}^x \underbrace{f_X(t)}_{\text{PDF: known}} dt, \quad x \in \mathbb{R}.$$

From the CDF to the PMF and PDF

What we are trying to answer

We know the CDF F_X . How do we recover the PMF p_X if X is discrete, or the PDF f_X if X is continuous?

What we know. The CDF is defined by

$$F_X(x) := P(X \leq x).$$

For every threshold x , it gives the probability of obtaining a value less than or equal to x . We want to recover the PMF or the PDF, depending on the type of X .

Discrete case: recover the PMF. The PMF is defined by

$$p_X(x) := P(X = x).$$

The CDF includes all values up to x . To isolate the probability of the single value x , we subtract the probability of values strictly less than x :

$$P(X = x) = P(X \leq x) - P(X < x).$$

Since the left limit satisfies

$$F_X(x^-) := \lim_{t \uparrow x} F_X(t) = P(X < x),$$

we obtain

$$\underbrace{p_X(x)}_{\text{PMF: to find}} = \underbrace{F_X(x) - F_X(x^-)}_{\text{CDF: known}}.$$

Thus the PMF at x is the size of the jump of the CDF at x .

Continuous case: recover the PDF. A PDF is a non-negative function f_X such that

$$P(a \leq X \leq b) = \int_a^b f_X(t) dt, \quad \int_{\mathbb{R}} f_X(t) dt = 1.$$

In particular,

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt.$$

The CDF is therefore the integral of the PDF. To recover a PDF, we differentiate the CDF:

$$\underbrace{f_X(x)}_{\text{PDF: to find}} = \frac{d}{dx} \underbrace{F_X(x)}_{\text{CDF: known}}.$$

This recovers a PDF at points where the derivative exists, by the fundamental theorem of calculus.